

Lecture 5

Contents of the lecture

- (1) HDL Role in Design Cycle
- (2) Simulation Tool
- (3) Synthesis Tool
- (4) Place & Route Tool



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HDL Role in Design Cycle

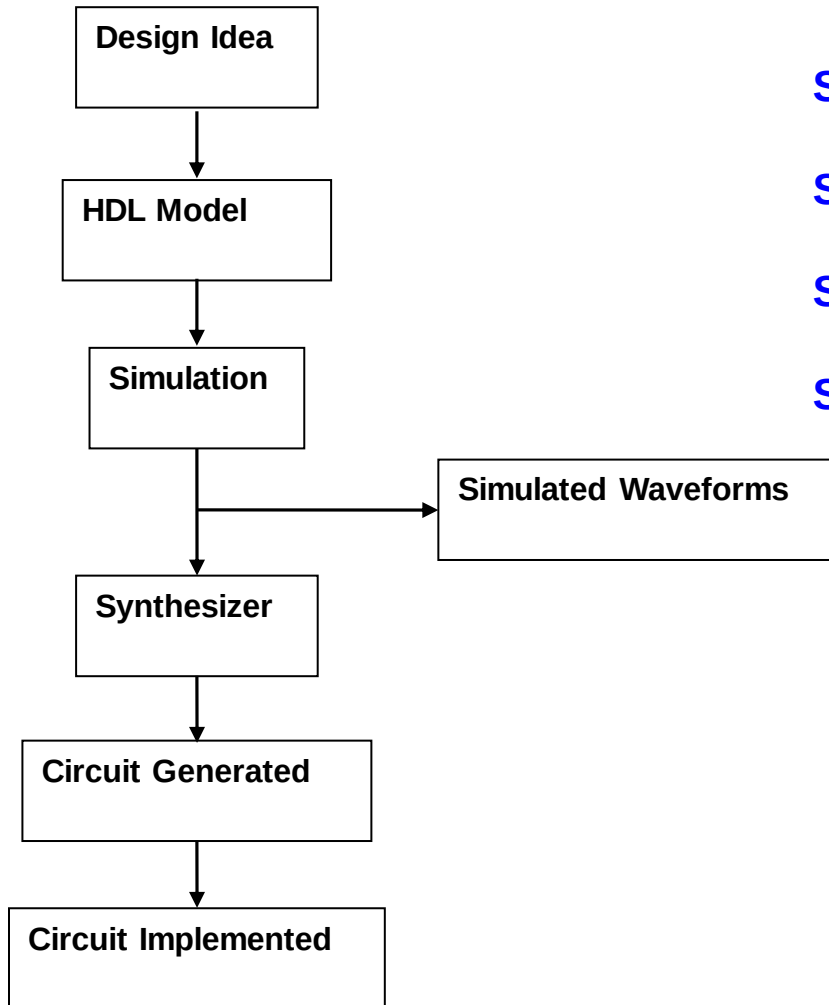
Step-1 : Write HDL Model

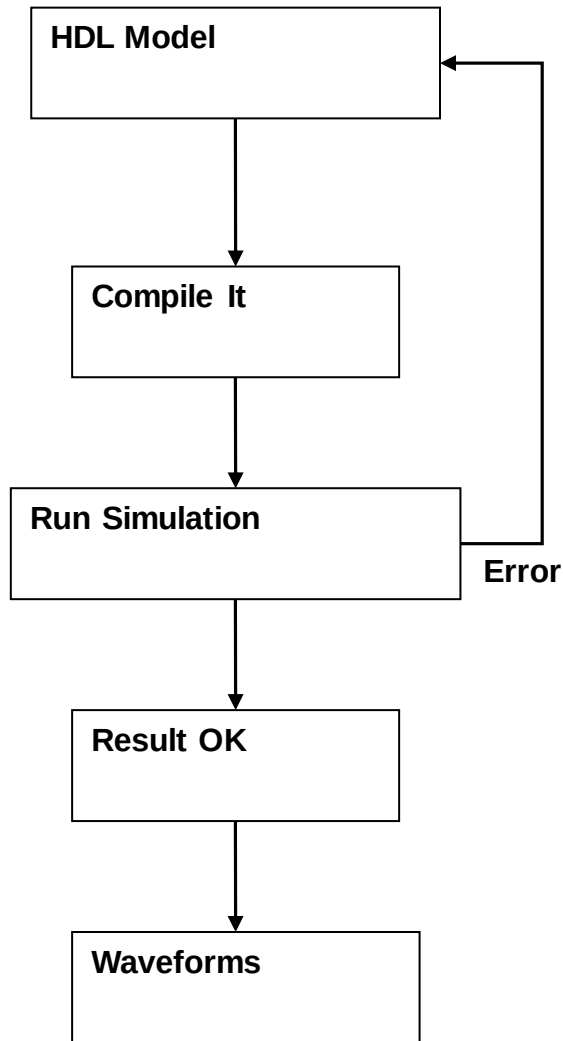
Step-2 : Check the Functionality

Step-3 : Synthesis the Design

Step-4 : Place & Route

Step-5 : Implement Design





```
module half_adder (a,b,s,c);
```

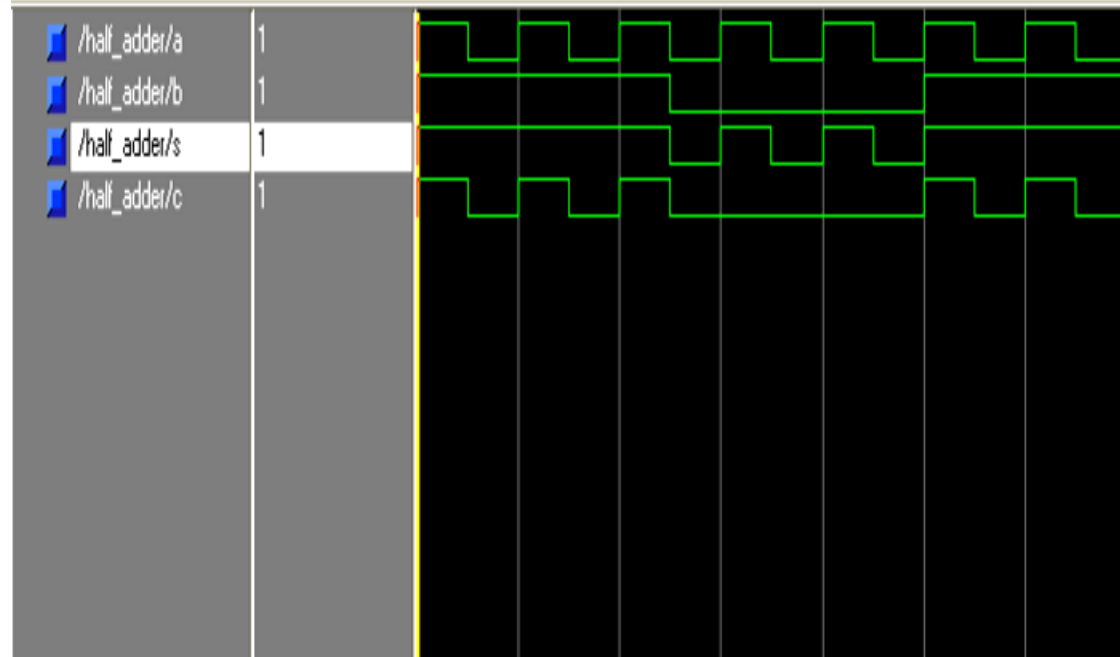
```
input a,b;
```

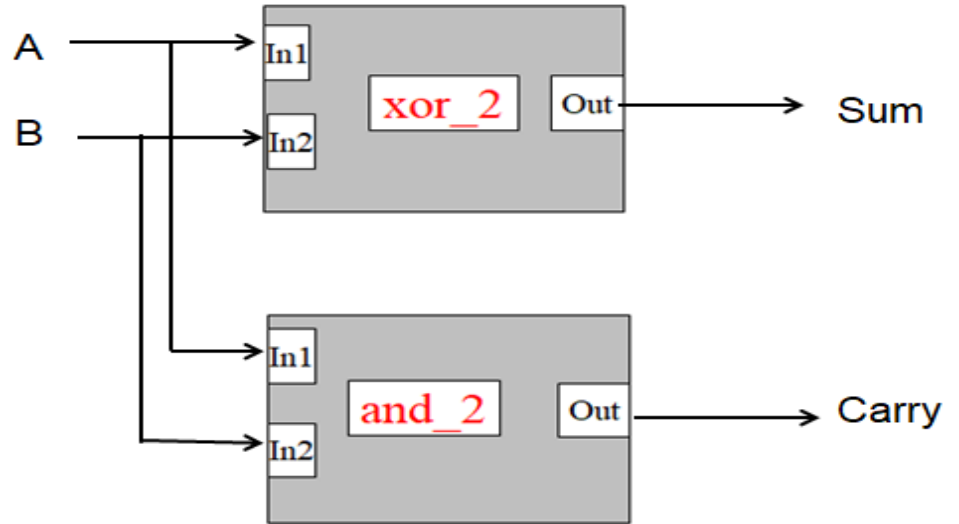
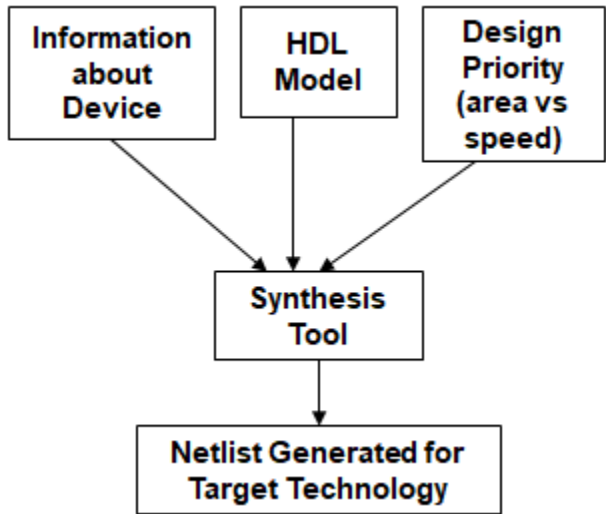
```
output s,c;
```

```
xor x1 (s,a,b);
```

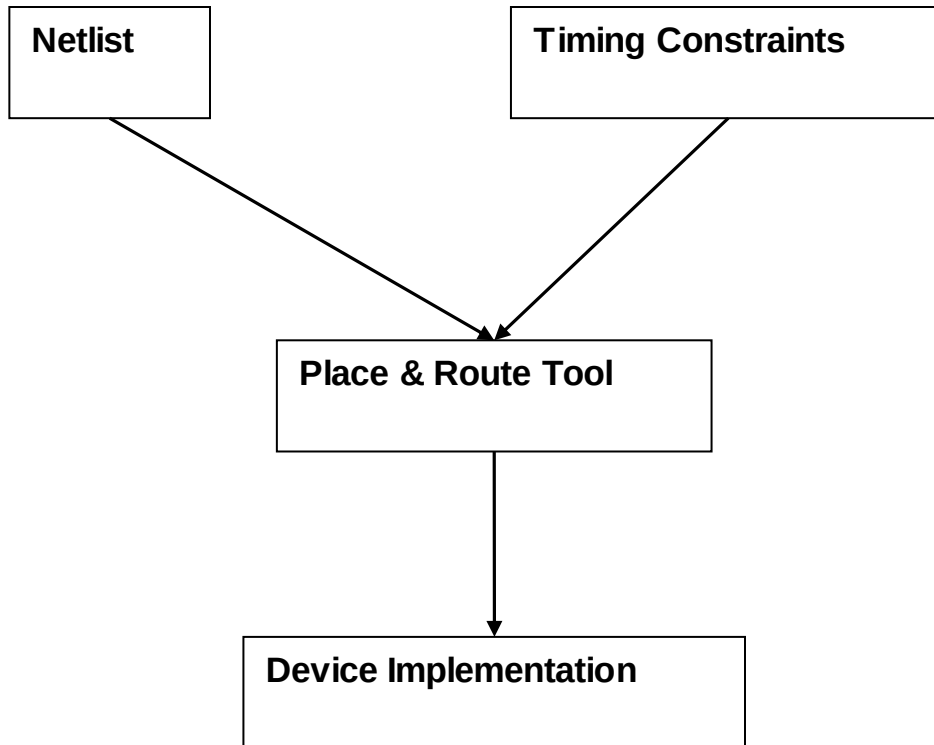
```
and a1 (c,a,b);
```

```
endmodule
```





Humility | Entrepreneurship | Teamwork and Relationships | Deliver the Promise | Learning | Social Responsibility | Respect for Individual



Rules:

- (i) Verilog is case sensitive
- (ii) Alphabet or an underscore can be used as the first character. Subsequent characters can be of **alphanumeric type, or the underscore (_), or the dollar (\$)** sign

Examples:

(i) Following are valid as identifiers:

- (a) name
- (b) _name
- (c) Name
- (d) name1
- (e) name_\$

(ii) Following are invalid as identifiers:

- (a) **\$name** not allowed as an identifier because of the presence of “\$” as the first character.
- (b) **1_name** not allowed as an identifier, since the numeral “1” is the first character
- (c) **@name** not allowed as an identifier because of the presence of the @ as the first character
- (d) **name aa** not allowed as an identifier because of the blank (“name” and “aa” are interpreted as two different identifiers)

Lecture 11

Contents of the lecture

- (1) Program Structure in Verilog
- (2) Level of abstraction
 - (A) Behavioural Level
 - (B) Dataflow level
 - (C) Structure Level
 - (D) Switch Level



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- (1) A program written in Verilog is called as 'module'
- (2) 'module' explains how a system is connected to outside world.
- (3) 'module' does not tells what the system is going to do or how the system is implemented

Syntax of module

```
module module_name(inputs,outputs);  
---  
---  
Logic of program  
---  
---  
endmodule
```

Declaration of inputs and outputs

(1) After declaration of module define input and output ports

Example:

```
input a,b; // a and b are two inputs ports of one bit each
```

Example:

```
output s; // s is output port of one bit
```

(2) If input or output is more than one bit then they can be defined as array

Example

```
input [3:0] a; // four bit input port a[3], a[2], a[1], a[0]
```

The level of abstraction means various methods by which we can write a verilog program.

(1) Behavioural Level

(2) Dataflow level

(3) Structure Level

(4) Switch Level

(1) Behavioural Level:

In this level of abstraction we explain the functionality of system using procedural and non-procedural statements which we find inside “always @” statement.

Example:

// behavioural model of two input and logic

```
module and_2_behavioural_level(a,b,y);  
input a,b;  
output y;  
always @(a,b)  
z=a & b;  
endmodule;
```

(2) Dataflow Level:

In this level of abstraction we explain the functionality of system using “assign” statements.

Example:

// dataflow model of two input and logic

```
module and_2_dataflow_level(a,b,y);  
input a,b;  
output y;  
assign z=a & b;  
endmodule
```

Structure Level (Gate Level):

- (i) In this level of abstraction we explain how the system is implemented rather than the functionality.
- (ii) In structure level we explain the system in terms of the interconnection of gates
- (iii) These gates are predefined in Verilog Library

and (output, inputs)
nand (output, inputs)
or (output, inputs)
nor (output, inputs)
xor (output, inputs)
xnor (output, inputs)
not (output, input)

Example:

// gate level model of two input and logic

```
module and_2_gate_level(a,b,y);  
input a,b;  
output y;  
and(c,a,b);  
endmodule;
```

(4) Switch Level of Modeling

- (i) In Verilog transistors are also referred as switches. These switch either in conducting mode (on state) or non conducting mode (off state)
- (ii) Verilog provides the ability to design at MOS transistor level.
- (iii) Designing at this level is rare due to more complexity of circuit (millions of transistor in a circuit).
- (iv) Keywords: nmos, pmos and cmos